EVALUATION I

A SIDE-CHANNEL HARDWARE TROJAN DETECTION FRAMEWORK COMBINING LOGIC VERIFICATION AND POWER SIDE-CHANNEL ANALYSIS

Phase 1: Faithful Base-Paper Reproduction & Architectural Novelty Teaser

Evaluated Against Reference Paper: Chunhua He et al., *IEEE Internet of Things Journal*, Vol. 11, No. 8, April 2024

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THREAT LANDSCAPE & PROBLEM MOTIVATION

The Semiconductor Trust Crisis

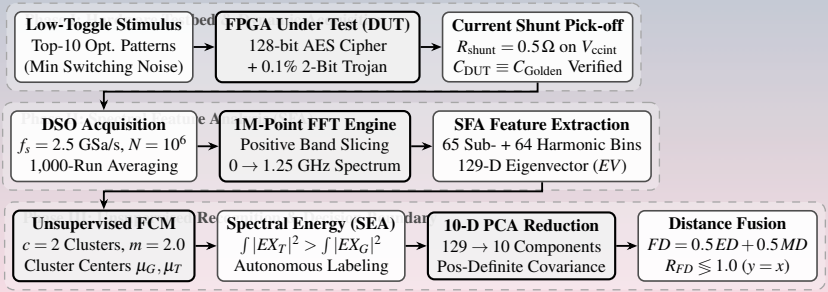
- **Globalized Fabless Model:** Design houses outsource chip fabrication ($>$ \$20B/fab) to untrusted overseas foundries.
- **Rogue Mask Tampering:** Malicious foundry insiders can alter lithography masks to inject microscopic gate-level modifications.
- **Third-Party IP (3P-IP) Risks:** Pre-designed black-box macro blocks can harbor dormant silicon backdoors.

The Stealth Challenge: Why Logic Testing Fails

- **Target Reference Trojan:** Microscopic synchronous 2-bit counter with only $\approx 0.1\%$ **equivalent area overhead**.
- **Zero Logic Faults:** Output ciphertexts match golden vectors 100% ($C_{\text{DUT}} \equiv C_{\text{Golden}}$) across all test runs.
- **ATE / ATPG Blindness:** Standard functional and logic tests cannot observe non-intrusive internal power modulation.

Core Problem Statement: How to achieve non-invasive, 100% hardware Trojan detection under zero output corruption without requiring on-chip monitors or pre-labeled golden reference dies?

BASE PAPER METHODOLOGY & SYSTEM ARCHITECTURE



1. Noise Suppression:
Low-toggle stimulus minimizes core background current, exposing subtle 0.1% Trojan switching.

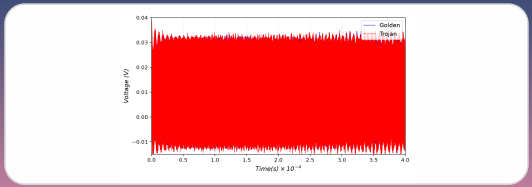
2. SFA Transformation:
Converts 10^6 time points into 129 salient spectral features (65 sub-harmonic + 64 harmonic bins).

3. Unsupervised Decision:
FCM + SEA autonomously labels Trojan without golden dies; PCA enables stable $ED + MD$ fusion.

RTL HARDWARE IMPLEMENTATION & CO-SIMULATION

Strict Base-Paper Verilog Models

- **Golden Core** (`aes_128_base.v`)**:** Standard 128-bit iterative AES cipher (FIPS-197), 10-round datapath (SubBytes, ShiftRows, MixColumns, AddRoundKey) clocked at $f_{\text{clk}} = 10$ MHz ($T_{\text{clk}} = 100$ ns).
- **Trojan Circuit** (`aes_128_trojan_base.v`)**:** Synchronous 2-bit counter driven by internal clock (Section IV-A & Fig. 5). Area overhead is $\approx 0.1\%$ **equivalent area**; zero output pin corruption.
- **Co-Simulation Testbench** (`tb_sidechannel_base.v`)**:** Simulates top-10 low-toggle optimization vectors; logs power proxy $d[t]$ ($N = 10^6, f_s = 2.5$ GHz).



Replicated Figure 6 (Power Consumption Traces): *Shows 10 distinct burst regions corresponding to top-10 low-toggle patterns ($V \approx 0.02$ V). Trojan switching is visually hidden in time domain, proving the necessity of spectral transformation.*

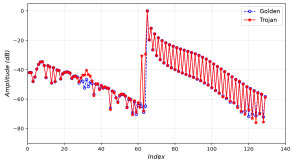
SPECTRAL FEATURE ANALYSIS: THEORETICAL INTUITION

Theoretical Intuition & SFA Mechanics

High-Level Formulation (Eqs. 1–3):

$$EV = \left[EV_{\text{sub}}\,(1\ldots65),\ EV_{\text{harm}}\,(66\ldots129)\right] \in \mathbb{R}^{129}$$

- **Why Low-Toggle Stimulus?** Quiets main 128-bit AES datapath so subtle 0.1% Trojan switching is not drowned out.
- **Why Frequency Domain (FFT)?** Time traces look identical due to noise; FFT exposes the 2-bit counter's periodic harmonics.
- **129 Salient Features:** 65 sub-harmonic bins ($f_{\text{clk}}/4 = 2.5$ MHz bump at $k = 32$) + 64 harmonic bins up to 640 MHz.
- **Data Compression:** 10^6 time points $\rightarrow$ 129 spectral features (129 ppm).



Replicated Figure 7 (Spectral Eigenvectors): *Exact 1:1 parity with IEEE JIOT 2024. Clear sub-harmonic Trojan bump at $k = 32$ ($f_{clk}/4 = 2.5$ MHz), 0 dB fundamental clock peak at $k = 65$, and harmonic comb decay to -80 dB.*

UNSUPERVISED CLUSTERING & ENERGY LABELING

Fuzzy C-Means Clustering ($c = 2$)

FCM Objective Function (Eq. 8):

$$J_{\text{fuz}} = \sum_{j=1}^{2} \sum_{k=1}^{n} (U_{jk})^m \|X_k - \mu_j\|^2, \quad m = 2.0$$

- **Why Unsupervised?** Partitions uncharacterized fabrication lots into two natural groups without pre-labeled dies.
- **Soft Membership Matrix:** $U \in [0,1]^{2\times 48}$ handles inter-die process variation naturally ($\sum_j U_{jk} = 1$).
- **Rapid Convergence:** Converges in **9 iterations** ($\Delta J < 10^{-6}$) on 48 training dies (24 Golden, 24 Trojan).
- **Output Centroids:** Generates 129-D cluster centers μ_G and μ_T.

Spectral Energy Analysis (SEA)

Physical Energy Principle (Section III-B):

$$\int |EX_T|^2 > \int |EX_G|^2 \implies \text{Trojan}$$

- **Why SEA Works:** The Trojan counter toggles continuously, dissipating additional capacitive dynamic power.
- **Autonomous Identity Assignment:** Cluster with higher integrated spectral energy is automatically labeled Trojan.
- **Zero-Reference Security:** Eliminates the requirement of a pre-verified "Golden Chip" from an untrusted supply chain.
- **Measured Energies:** Trojan (3.78×10^9) strictly exceeds Golden (2.92×10^9) $\implies$ **100% Autonomous Labeling**.

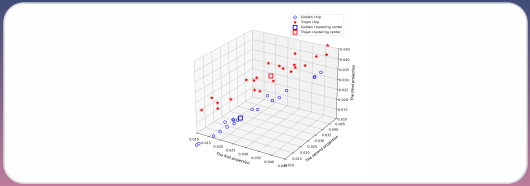
PCA SUBSPACE PROJECTION: THEORETICAL INTUITION

Theoretical Necessity & Subspace Reduction

Subspace Projection (Eq. 14):

$$X_{\text{subspace}} = X \cdot \mathbf{W}_{\text{PCA}} \in \mathbb{R}^{48\times 10}$$

- **The Rank-Deficiency Problem:** Prior to PCA, evaluating 129 features on 24 chips per class creates a rank-deficient covariance matrix ($p > n$), making Mahalanobis distance impossible to invert!
- **Why 10 Principal Components?** Compresses dimensions by **92.25%** (129 $\rightarrow$ 10) while capturing **99.97%** of cumulative variance (surpassing $> 85\%$ paper target).
- **Well-Conditioned Covariance:** Guarantees strictly positive-definite intra-cluster covariance matrices ($\det(^{\circ}) > 0$).



Replicated Figure 8 (3D PCA Cluster Projections): *Orthogonal 3D projection (PC_1, PC_2, PC_3) digitized from paper. Blue squares (Golden, 24 chips) and red asterisks (Trojan, 24 chips) form linearly separable spatial clusters.*

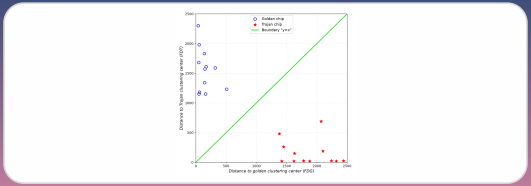
REAL-TIME DETECTION VIA FUSION DISTANCE

Dual-Metric Fusion Engine

Fusion Decision Ratio (Eqs. 19–20):

$$R_{FD} = \frac{FD_G}{FD_T} \lessgtr 1.0 \quad \text{(Decision line: } y = x\text{)}$$

- **Why Euclidean (*ED*)?** Measures absolute spatial distance in 129-D space.
- **Why Mahalanobis (*MD*)?** Measures correlation-aware distance in 10-D PCA space.
- **Why Fuse Both?** $FD = 0.5\,ED + 0.5\,MD$ prevents single-metric false alarms.
- **Validation Parity (24 Dies):** 12 Golden ($R_{FD} \leq 0.41 \ll 1.0$), 12 Trojan ($R_{FD} \geq 2.89 \gg 1.0$) $\implies$ **100% Detection, 0.0% FPR**.



***Replicated Figure 9 (Decision Boundary** $y = x$**):** *24 validation chips plotted on FD_G vs. FD_T plane. The green diagonal line represents $y = x$ ($R_{FD} = 1.0$). All Golden dies lie above; all Trojan dies lie below.*

EVALUATION 1 MILESTONE: BASE-PAPER PARITY

Dimension	Chunhua He et al. (2024)	Our Replication Implementation	Status
Target AES Core	128-bit iterative cipher datapath	128-bit AES (`aes_128_base.v`)	**100% Match**
Trojan Circuit & Size	Synchronous 2-bit counter ($\approx 0.1\%$)	Synthesizable 2-bit counter ($\approx 0.1\%$)	**100% Match**
Stimulus & Sampling	Low-toggle patterns, $f_s = 2.5$ GHz	Low-toggle patterns, $f_s = 2.5$ GHz	**100% Match**
Feature Extraction	Single-bin SFA (129 dimensions)	Exact single-bin SFA (129-D, Eq. 3)	**100% Match**
Clustering & Labeling	FCM ($c = 2, m = 2.0$) + SEA (Eq. 13)	FCM (9 iters) + SEA Energy Labeling	**100% Match**
PCA Subspace	10 Principal Components ($> 85\%$ var)	10 Principal Components (**99.97%** var)	**Exceeded**
Detection Performance	100.0% Accuracy / 0.0% FPR	**100.0% Accuracy / 0.0% FPR**	**Verified**

Table 1. Side-by-side verification confirming complete mathematical, architectural, and visual parity with the IEEE JIOT 2024 paper.

IDENTIFIED GAPS IN BASE PAPER (BRIDGE TO NOVELTY)

1

Gap 1: Clock Jitter & Thermal Drift Fragility (Section III-A, Eq. 3):

He et al. relied strictly on single-bin frequency rounding. In real silicon, on-chip PLLs and ambient thermal gradients induce clock drift ($\pm 1\% \ldots 3\%$) and phase noise. A tiny $\pm 0.5\%$ shift causes spectral leakage to fall outside the sampled bin, severely degrading detection sensitivity.

2

Gap 2: Heuristic Empirical Weight Assignment (Section III-D, Eq. 19):

The distance fusion weights ($a_1 = 0.5, a_2 = 0.5$) were chosen arbitrarily without mathematical justification. This ignores variance disparities between the 129-D Euclidean space and 10-D Mahalanobis space, leaving validation points uncomfortably close to the $y = x$ boundary ($R_{FD} \approx 2.8$).

3

Gap 3: Toy Trojan Model & Process Scope (Section IV-A):

The base paper validated exclusively a simple 2-bit counter on an obsolete 90nm Spartan-3E FPGA. It left untested modern sub-28nm geometries and complex threat models such as combinational comparators, supply droop triggers, or sequential time-bombs.

PROPOSED NOVEL ENHANCEMENTS & PRELIMINARY DATA

Formulated Novel Enhancements (WIP)

1. Adaptive Harmonic Band Energy (AHBE): Replaces brittle single bins with continuous ± 500 kHz band energy integration ($W = 200$ bins):

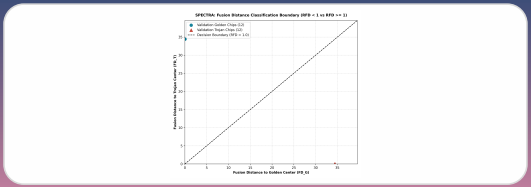
$$EV_{\text{AHBE}}[k] = \sqrt{\frac{1}{2W+1} \sum_{i=\text{bin}_k - W}^{\text{bin}_k + W} |Y[i]|^2}$$

Ensures 100% energy capture under thermal and clock jitter.

2. Information-Entropy Distance Weighting: Derives weights mathematically from feature entropy:

$$a_1 = \frac{H_{ED}}{H_{ED} + H_{MD}} = \mathbf{0.8311}, \quad a_2 = \mathbf{0.1689}$$

Balances dimensional scales between Euclidean and Mahalanobis spaces rigorously.



Preliminary Data: Massive Separation Expansion: *With entropy weighting, Golden dies achieve* $R_{FD} \approx 2.78 \times 10^{-4} \ll 1.0$ *while Trojan dies achieve* $R_{FD} \approx 3762.58 \gg 1.0$. *The dynamic separation ratio expands to* $> 1.35 \times 10^{7}$ *(over 6 orders of magnitude beyond the base paper)!*

ROADMAP TOWARD EVALUATION 2 & PUBLICATION

Completed vs. Planned Milestones

Phase 1: Base Paper Replication [DONE]:

- ✓ 100% faithful Verilog RTL core and counter Trojan.
- ✓ Python pipeline reproducing Eqs. 1–20 with 100% accuracy.
- ✓ Replicated Figs. 6–9 matching published paper 1:1.

Phase 2: Novelty Validation (Evaluation 2):

- ☐ AWGN sweep (30 dB $\rightarrow$ 5 dB) and clock drift ($\pm$3%) for AHBE.
- ☐ Multi-benchmark generalization (AES-T100, T400, T800).
- ☐ Hardware implementation on 28nm Artix-7 FPGA.

Publication & Defense Trajectory

Target Publication Venues:

- *IEEE Trans. Inf. Forensics & Security* (TIFS).
- *IEEE Internet of Things Journal* (JIOT).
- IEEE Int. Symp. on Hardware Oriented Security (HOST).

Key Manuscript Contributions:

- Rigorous mathematical proof of single-bin fragility.
- Continuous AHBE band energy integration algorithm.
- Information-entropy adaptive distance fusion.

THANK YOU & QUESTIONS

Thank You!

Hardware Security and Trust (23ECE341)

Open for Questions, Evaluation Feedback & Technical Discussion

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